

Project Design Phase-II Technology Stack (Architecture & Stack)

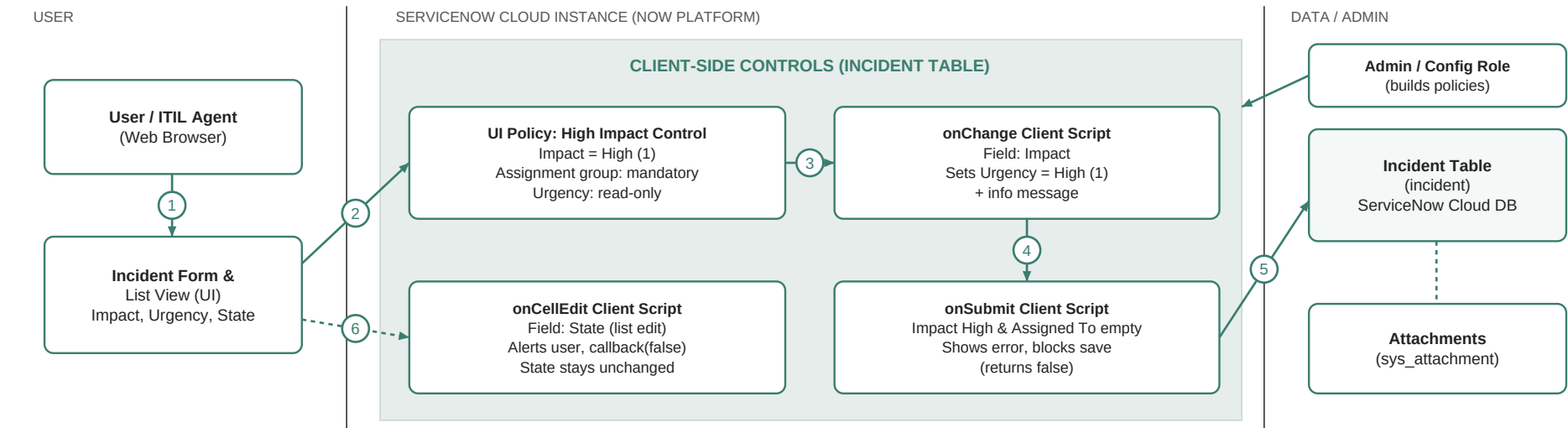
Date	31 January 3035
Team ID	6ab75aba534c8646a9ce4107
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Implement Client Script & UI Policy (Incident) – enforcing Incident data integrity in ServiceNow

Table-1 : Components & Technologies:



S.No	Component	Description	Technology
1.	User Interface	How user interacts with application: ITIL users and agents create and update Incidents through the ServiceNow Incident form and Incident list view in a web browser.	ServiceNow Platform UI (Core UI), HTML, CSS, JavaScript

2.	Application Logic-1	UI Policy “High Impact Control” (table: Incident; condition: Impact is 1 – High). Makes Assignment group mandatory and Urgency read-only; Reverse if false = true restores fields when Impact changes.	ServiceNow UI Policy & UI Policy Actions
3.	Application Logic-2	onChange Client Script “Auto set urgency for high impact” on field Impact. When Impact = 1, sets Urgency = 1 and displays an info message.	ServiceNow Client Script (onChange), JavaScript, g_form API
4.	Application Logic-3	onSubmit Client Script blocks save when Impact is High and Assigned To is empty. onCellEdit Client Script blocks State changes made from the list view.	ServiceNow Client Script (onSubmit, onCellEdit), JavaScript
5.	Database	Incident records (table: incident) store Impact, Urgency, Assignment group, Assigned To and State values validated by the client-side rules.	ServiceNow relational database (Incident table)
6.	Cloud Database	Managed database service hosted with the ServiceNow cloud instance; no separate database setup needed.	ServiceNow Cloud (Now Platform) managed database
7.	File Storage	Optional attachments on Incident records are stored in the platform attachment tables.	ServiceNow Attachment storage (sys_attachment)
8.	External API-1	No third-party API is required. Optional integration with other systems can use the platform’s built-in REST interface.	ServiceNow Table REST API (optional)
9.	External API-2	Not used in this project.	Not applicable
10.	Machine Learning Model	Not used. Data integrity is enforced through rule-based UI Policies and Client Scripts.	Not applicable
11.	Infrastructure (Server / Cloud)	Application Deployment on Cloud (ServiceNow instance) Local Server Configuration: None; only a modern web browser is needed. Cloud Server Configuration: ServiceNow-hosted instance; admin/configuration role to create policies and scripts.	ServiceNow Cloud (Now Platform) instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open web standards used for client-side logic and UI; the ServiceNow platform itself is proprietary.	JavaScript, HTML, CSS
2.	Security Implementations	Role-based access control and ACLs, admin-only configuration, HTTPS/TLS access, mandatory-field and save-time validation to prevent incomplete data.	ServiceNow ACLs, RBAC, TLS (HTTPS)
3.	Scalable Architecture	Multi-tier platform (browser, application nodes, database). Validation runs in the user’s browser, and one policy applies to every Incident form.	ServiceNow multi-tier Now Platform

S.No	Characteristics	Description	Technology
4.	Availability	Cloud-hosted instance with redundant infrastructure managed by ServiceNow; the logic is stored on the platform, so it is available whenever the instance is.	ServiceNow cloud data centers (managed failover)
5.	Performance	Lightweight scripts run only on specific events (Impact change, save, list edit), so there are no extra server round trips for validation.	Client-side execution, browser caching

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>